

FORGE · PRODUCTION TRACKING PLATFORM

Requirements Document

Business objectives, scope, stakeholders, functional and non-functional requirements, and the technology decision register for the Forge production tracking platform.

INTERNAL

PREPARED FOR

Symbiosys Technologies

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SYM-FRG-REQ-001

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Deployed and in production use

AUTHOR

Product & Engineering

0.1 Document control

Revision history

VERSION	DATE	AUTHOR	SUMMARY OF CHANGE
0.1	Sep 2026	Product & Engineering	Initial structure, stakeholder analysis, scope definition.
1.0	Sep 2026	Product & Engineering	First complete draft: full requirement set, process models, and technology decision register.

Approval

ROLE	NAME	SIGNATURE	DATE
Project Sponsor			
Head of Product			
Engineering Lead			
Production Head (user representative)			

Related documents

ID	DOCUMENT	RELATIONSHIP
SYM-FRG-SRS-001	Software Requirements Specification	Refines this document into testable software requirements
SYM-FRG-USG-001	Product Usage & User Guide	Documents the delivered system
SYM-FRG-MKT-001	Marketing Analysis	Consumes proven capability from this document

REQUIREMENT IDENTIFIER SCHEME

Used consistently across this document and the SRS so that traceability holds in both directions.

- **BR-*nnn*** — business requirement, stated in business language in this document.
- **FR-MODULE-*nnn*** — functional requirement, refined in the SRS.
- **NFR-CAT-*nnn*** — non-functional requirement.
- **DEC-*nnn*** — technology or architecture decision record (Section 10).

0.2 Contents

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1 Executive summary

Forge is a production tracking and review platform built for Symbiosys Technologies' animation and VFX operation, replacing spreadsheet-based shot tracking and ad-hoc review with a single system of record.

1.1 The problem

Production tracking ran on spreadsheets maintained per show. Shot status was accurate only as of the last manual update, review feedback arrived through chat and email where it was frequently lost, and approvals left no record of who signed off on what. Commercial alternatives were evaluated and rejected on cost, on their cloud-hosted delivery model, and on the pipeline work required to adapt them to the studio's existing episodic tracksheet format.

1.2 The solution

A self-hosted web platform covering the full production path: import of existing tracksheets, hierarchical tracking from project through episode, sequence and shot to individual task, role-scoped task assignment, a frame-accurate review player with annotation, a multi-stage approval chain, external client review by access link, and attendance and effort recording.

1.3 Current status

The system is deployed and in production use. At the time of writing it holds 1,065 shots and 1,066 tasks imported from live production tracksheets, across 65 user accounts spanning six roles. This document therefore records requirements for a system that largely exists — it is both a specification and a record of what was built, and each requirement carries a delivery status.

1.4 Requirement summary

CATEGORY	MUST	SHOULD	COULD	TOTAL
Functional requirements	48	14	9	71
Non-functional requirements	19	7	3	29
Business requirements	11	3	2	16

Table 1.1 — Requirement counts by priority. Full listings in Sections 8 and 9.

2 Business background and problem statement

2.1 Operating context

Symbiosys Technologies produces animation and visual effects work, principally episodic content organised into episodes, sequences and shots. Work passes through departmental stages, each contributing to a shot before it advances. A single episode commonly contains several hundred shots, each requiring individual tracking, review and sign-off.

2.2 Problems with the previous approach

PROBLEM	OBSERVED CONSEQUENCE	COST TO THE BUSINESS
Status held in spreadsheets	Accurate only as of the last manual edit; concurrent editing produced conflicting copies.	Production meetings spent establishing current state rather than acting on it.
Review feedback in chat and email	Notes detached from the frame and the version they referred to; lost when threads scrolled.	Avoidable retakes; artist time spent on work that had already been corrected.
No approval record	No reliable answer to who approved a shot, when, or on what version.	Disputes with clients over what was signed off; no audit trail.
Client delivery ad hoc	Media sent by file transfer, feedback returned by email in free text.	Slow review rounds; feedback not attached to the shot record.
No effort data	Time per shot unrecorded or reconstructed from memory.	Bids based on estimate rather than history.
Commercial tools rejected	Per-seat licensing at studio headcount; cloud hosting of unreleased client media; configuration effort to match the existing tracksheet structure.	Either unaffordable cost or unacceptable delivery model.

Table 2.1 — Problem statement with observed consequence and business cost.

2.3 Business requirements

ID	REQUIREMENT	PRIORITY
BR-001	The studio shall maintain a single authoritative record of the status of every shot in production.	MUST
BR-002	Review feedback shall be permanently associated with the specific version and frame it refers to.	MUST
BR-003	Every approval decision shall be recorded with its actor, timestamp and subject version.	MUST
BR-004	Existing production tracksheets shall be importable without reformatting or re-entry.	MUST
BR-005	All production media shall remain on infrastructure controlled by the studio.	MUST
BR-006	The system shall carry no per-user licensing cost.	MUST
BR-007	Access to production data shall be restricted according to the user's role and department.	MUST
BR-008	External clients shall be able to review and comment without holding a studio account.	MUST
BR-009	Effort expended per shot shall be recorded to inform future bidding.	MUST
BR-010	The system shall operate on the studio's existing hardware without internet dependency for core function.	MUST
BR-011	Administrative oversight shall be separable from production authority.	MUST
BR-012	The system shall support onboarding of new staff without administrator intervention for every account, while reserving the granting of authority to an administrator.	MUST
BR-013	Historical changes should be reversible where a mistake is identified after the fact.	SHOULD
BR-014	The system should provide studio-wide visibility of workload distribution.	SHOULD
BR-015	The platform could be extended by third-party plug-ins.	COULD
BR-016	The platform could integrate directly with content creation applications.	COULD

Table 2.2 — Business requirements. Each is traced to functional requirements in Appendix B.

3 Business objectives and success criteria

Each objective carries a measure and a baseline. Objectives without a measure are excluded, since they cannot be used to determine whether the project succeeded.

ID	OBJECTIVE	MEASURE	BASELINE AND TARGET
OBJ-1	Establish a single source of truth for shot status	Proportion of active shots tracked in the system	Baseline 0%. Target 100% of shots on active shows.
OBJ-2	Eliminate per-seat licensing cost	Annual licensing spend on production tracking	Baseline: quoted per-seat cost at 65 accounts. Target: nil.
OBJ-3	Retain review feedback permanently against its version	Proportion of review notes recorded against a version and frame	Baseline effectively 0% (chat and email). Target 100%.
OBJ-4	Make approvals auditable	Time to establish who approved a given shot version	Baseline: manual search of message history, frequently unresolvable. Target: single screen.
OBJ-5	Adopt without production disruption	Re-keyed rows during migration	Baseline: full re-entry under alternatives. Target: zero.
OBJ-6	Keep client media on studio infrastructure	Media assets stored outside studio control	Baseline: transfers to third-party services. Target: nil.
OBJ-7	Record effort against shots	Proportion of completed tasks with logged time	Baseline 0%. Target 90% of completed tasks.

Table 3.1 — Business objectives with measures. Traced to business requirements in Appendix B.

4 Scope

4.1 In scope

AREA	INCLUDED
Identity and access	Authentication, self-service registration, password recovery, session management, role-based access control, capability administration.
Production structure	Projects, episodes, sequences, shots, assets, departments, and configurable pipeline stages.
Tracking	Global tracking grid, filtering and grouping, saved views, spreadsheet import, CSV export.
Task management	Creation, assignment, claiming, status progression, dependencies, checklists, comments, attachments.
Review	Version management, media upload, frame-accurate playback, annotation, comparison tooling, frame-anchored comments, presentation mode.
Approval	Multi-stage approval chain with recorded history and change requests.
Client delivery	Access links with codes, external review interface, client feedback capture, delivery records.
Time and attendance	Automatic clock-in on sign-in, presence, daily logs, timesheets, daily standup.
Communication	Notifications, team chat, studio broadcasts.
Administration	User and role administration, studio settings, audit log with rollback, schema and workflow configuration.
Operations	Containerised deployment, database migration on start-up, scheduled backup.

Table 4.1 — In-scope functional areas.

4.2 Out of scope

Explicitly excluded from this phase. Listing these prevents later assumption that they were implied.

EXCLUDED	RATIONALE
DCC application integration Maya, Blender, Nuke, Houdini plug-ins	Substantial per-application engineering effort. Deferred; the administrative surface exists but is gated and marked as forthcoming.
Render farm submission and monitoring	Requires integration with farm management software not yet selected.
Mobile and desktop applications	Scaffolding present in the codebase; not built, tested or released.
Managed cloud hosting	Contradicts BR-005 and BR-010 for the primary deployment.
Automated media transcoding	No transcode pipeline; media is played as uploaded.
Financial accounting integration	Financial views are reporting only; no ledger integration.
Multi-studio federation	The data model is multi-tenant, but cross-tenant sharing is not implemented.

Table 4.2 — Out of scope, with rationale.

5 Stakeholders and user classes

Six roles, each defined by what it is accountable for and — equally important — what it is deliberately prevented from doing. The separation of administrative oversight from production authority (BR-011) is a designed constraint, not an omission.

ROLE	ACCOUNTABLE FOR	CAN	CANNOT
Artist	Completing assigned shot work	View own work, update task status, log time, upload versions, annotate, submit for review, comment	Assign work to others, approve any stage, view other departments' data
Lead	Quality and throughput of one department	Everything an artist can, plus create and assign tasks within the department, approve at lead stage, request changes	Act outside own department, give final production sign-off
Production Head	Studio-wide delivery	Studio-wide visibility, task creation and assignment anywhere, final approval, client delivery, user administration	—
Producer	Final gate before client delivery	All production authority studio-wide, plus administrative capability	—
Administrator	Standing the studio up and monitoring it	Manage accounts, roles, licences, integrations, pipeline configuration; view financial data; observe all activity; comment for oversight	Create, assign, edit or be assigned tasks; submit or approve reviews; contact clients
Client external	Reviewing delivered work	View shared versions via access link and code, submit feedback, approve or request changes on shared items	Access the studio application, view anything not explicitly shared

Table 5.1 — User classes with authority and constraint.

ON THE ADMINISTRATOR CONSTRAINT

The administrator role holds no task or review capability. This follows from BR-011: the person who administers accounts and monitors the studio should not also be a participant in the production approval chain, so that oversight remains independent of the work being overseen. The constraint is enforced server-side by capability grant, not by hiding controls in the interface.

5.2 Other stakeholders

STAKEHOLDER	INTEREST	INFLUENCE ON REQUIREMENTS
Project sponsor	Cost, delivery, business outcome	Approves scope and objectives
IT / infrastructure	Deployment, backup, security	Constrains hosting and operational requirements
Client organisations	Review experience, media security	Drives BR-005 and BR-008
Engineering team	Maintainability, technical debt	Drives Section 10 decisions

Table 5.2 — Non-user stakeholders.

6 System context

The boundary between Forge and everything around it: who uses it, what it depends on, and what crosses the studio's network perimeter.

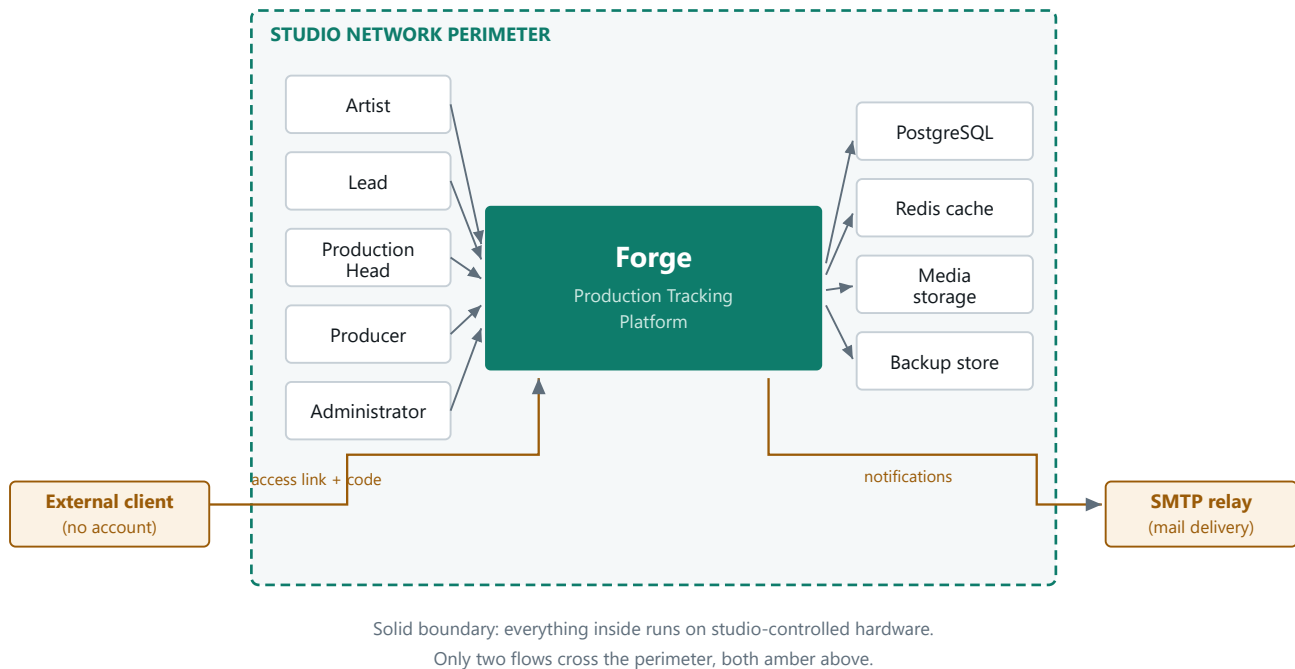


Figure 6.1 — System context. Production media never crosses the studio perimeter, satisfying BR-005. The only outbound dependency is mail delivery; the only inbound is client review by access link.

7 Current state and proposed state

7.1 Current-state process

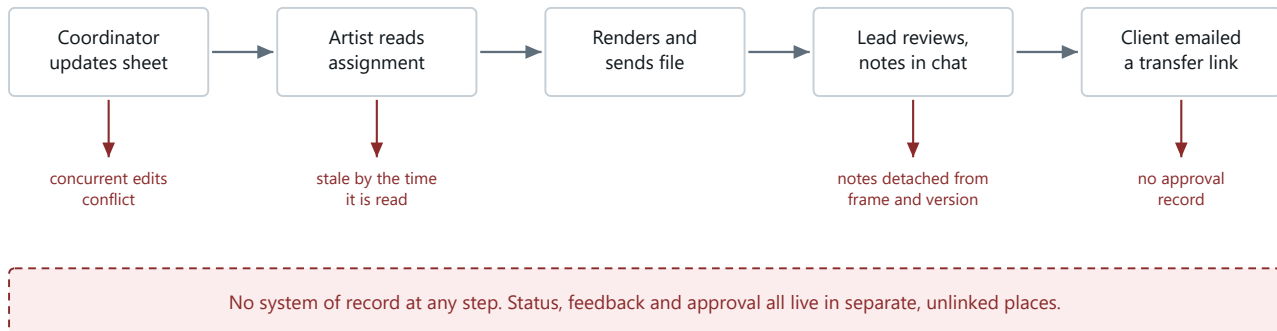


Figure 7.1 — Current-state process, with the four failure points identified in Table 2.1.

7.2 Proposed-state process

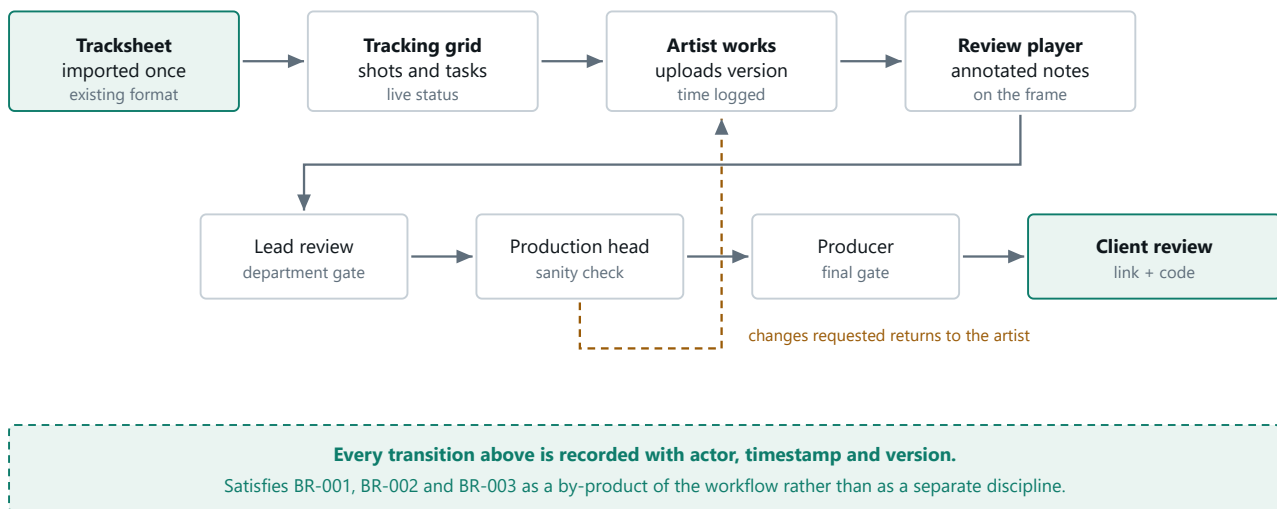
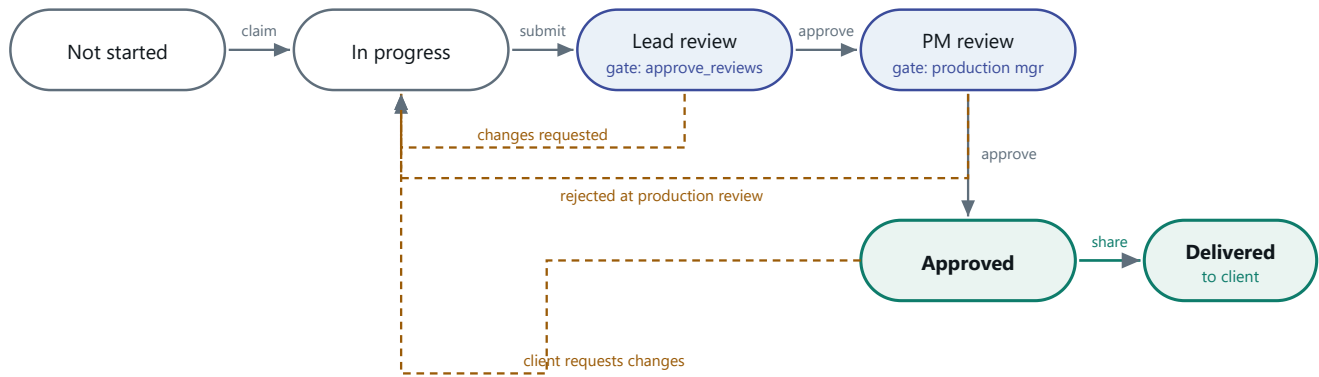


Figure 7.2 — Proposed-state process as implemented. The dashed amber path is the change-request loop returning work to the artist.

7.3 Approval chain state model

The formal state machine behind Figure 7.2. Every transition is capability-gated; the gates are specified in Section 8.6 and enforced server-side.



Every transition writes an approval event: actor, action, timestamp, and the version it applied to.
A rejection at any stage returns the task to In progress; it does not skip backwards one stage at a time.

Figure 7.3 — Approval chain state model. Solid transitions advance; dashed amber transitions return work to the artist.

7.4 Production data hierarchy

The structural backbone of the system. The hierarchy mirrors how episodic production is actually organised, which is what makes tracksheet import possible without a mapping step.

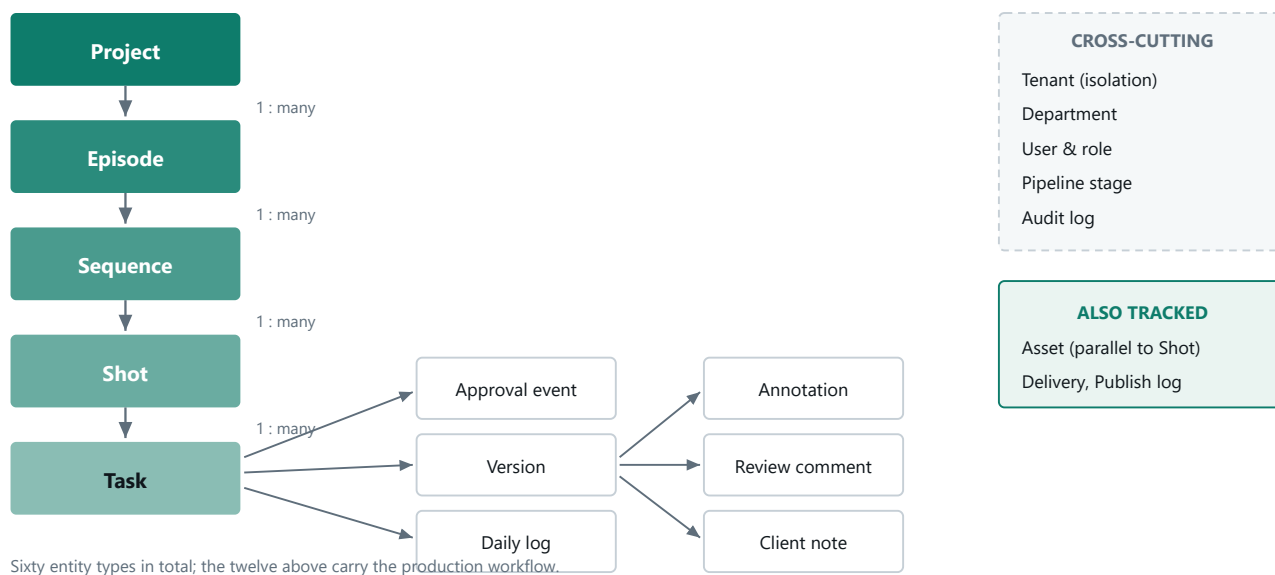


Figure 7.4 — Production data hierarchy. Assets follow a parallel path to shots for non-shot deliverables such as characters, props and environments.

7.5 Tracksheet import flow

The mechanism that satisfies BR-004 and removes the principal adoption barrier. Its design principle is tolerance: accept the studio's file as it is, and never discard data that is not understood.

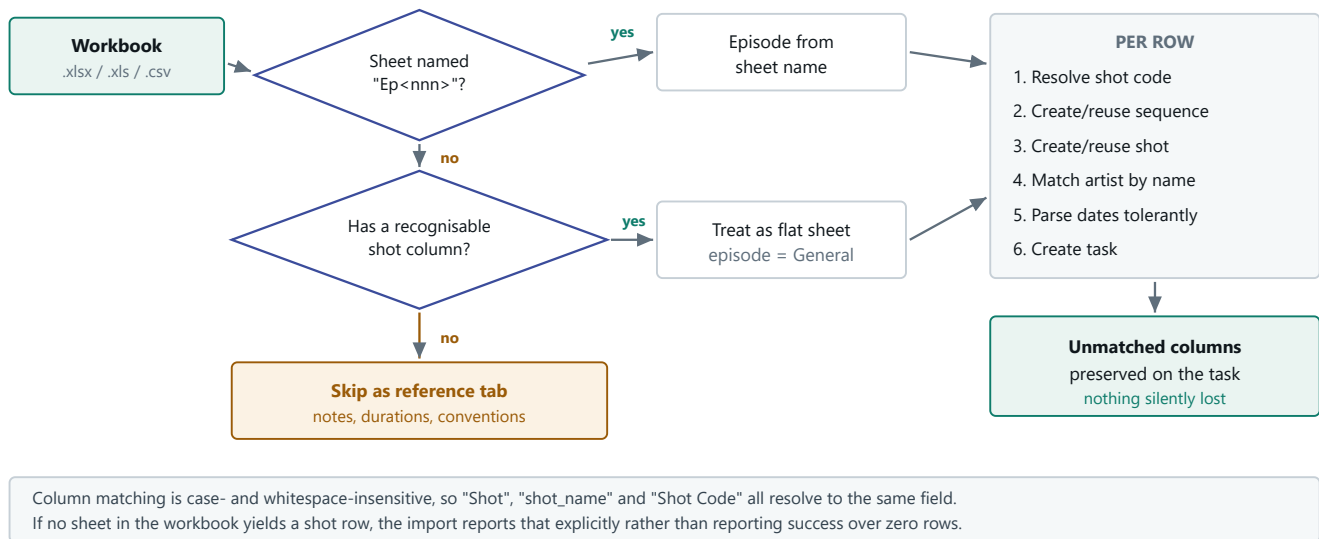


Figure 7.5 — Tracksheet import decision flow. Diamonds are decisions; the amber path discards only sheets proven to contain no shot data.

8 Functional requirements

Grouped by module. Each requirement carries a priority and a delivery status. Requirements marked **DELIVERED** are implemented and in production use; the SRS refines each into testable form.

8.1 Authentication and identity

ID	REQUIREMENT	PRI.	STATUS
FR-AUTH-001	Users shall authenticate with an email address and password.	MUST	DELIVERED
FR-AUTH-002	Passwords shall be stored only as irreversible hashes.	MUST	DELIVERED
FR-AUTH-003	New staff shall be able to register an account without administrator action.	MUST	DELIVERED
FR-AUTH-004	Users shall be able to recover access by email-delivered reset link.	MUST	DELIVERED
FR-AUTH-005	Reset tokens shall be single-use and time-limited.	MUST	DELIVERED
FR-AUTH-006	Users shall be able to change their own password when authenticated.	MUST	DELIVERED
FR-AUTH-007	Sessions shall be invalidated on logout, password change, role change, and deactivation.	MUST	DELIVERED
FR-AUTH-008	Deactivated accounts shall be refused access while retaining all their historical records.	MUST	DELIVERED
FR-AUTH-009	Repeated failed authentication attempts shall be rate limited.	MUST	DELIVERED
FR-AUTH-010	The registration form shall collect the department the new starter will work in, so their lead can find and assign them without administrator intervention.	MUST	DELIVERED
FR-AUTH-011	The registration form shall let a new starter state the role they are joining as.	SHOULD	DELIVERED
FR-AUTH-012	A role above the least-privilege role shall be recorded as a request and granted only by an administrator; self-registration shall never confer elevated authority.	MUST	DELIVERED
FR-AUTH-013	Accounts awaiting a role decision shall be identifiable in the studio roster.	SHOULD	DELIVERED

Table 8.1 — Authentication and identity requirements.

8.2 Access control

ID	REQUIREMENT	PRI.	STATUS
FR-RBAC-001	Every action shall be authorised against a named capability rather than a hard-coded role check.	MUST	DELIVERED
FR-RBAC-002	Capability grants shall be held in the database and administrable at runtime.	MUST	DELIVERED
FR-RBAC-003	Authorisation shall be enforced server-side; interface concealment alone shall not constitute enforcement.	MUST	DELIVERED
FR-RBAC-004	The administrator role shall hold no task or review capability.	MUST	DELIVERED
FR-RBAC-005	Data visibility shall be scoped as studio-wide, department, or own work according to role.	MUST	DELIVERED
FR-RBAC-006	All data access shall be isolated by tenant.	MUST	DELIVERED
FR-RBAC-007	Controls the current user cannot exercise shall not be displayed to them.	SHOULD	DELIVERED
FR-RBAC-008	Direct navigation to a page outside the user's role shall be refused.	MUST	DELIVERED

Table 8.2 — Access control requirements.

8.3 Production structure and tracking

ID	REQUIREMENT	PRI.	STATUS
FR-TRK-001	The system shall represent projects, episodes, sequences, shots and assets in a hierarchy.	MUST	DELIVERED
FR-TRK-002	A global tracking grid shall present every shot with its status, assignee and review state.	MUST	DELIVERED
FR-TRK-003	The grid shall support filtering by project, episode, sequence, artist, position and department.	MUST	DELIVERED
FR-TRK-004	Status shall be editable directly in the grid.	MUST	DELIVERED
FR-TRK-005	Grid configurations shall be saveable as named views.	SHOULD	DELIVERED
FR-TRK-006	The grid shall be exportable to CSV.	SHOULD	DELIVERED
FR-TRK-007	Departments shall be configurable with an ordered pipeline position.	MUST	DELIVERED

Table 8.3 — Tracking requirements.

8.4 Tracksheet import

ID	REQUIREMENT	PRI.	STATUS
FR-IMP-001	The system shall import spreadsheet tracksheets in .xlsx, .xls and .csv formats.	MUST	DELIVERED
FR-IMP-002	Import shall accept multi-sheet workbooks organised one sheet per episode.	MUST	DELIVERED
FR-IMP-003	Import shall accept a single flat sheet containing all rows.	MUST	DELIVERED
FR-IMP-004	Column names shall be matched case- and whitespace-insensitively across known synonyms.	MUST	DELIVERED
FR-IMP-005	Sheets containing no recognisable shot data shall be skipped, not imported as invalid rows.	MUST	DELIVERED
FR-IMP-006	Columns the importer does not recognise shall be preserved against the created task.	MUST	DELIVERED
FR-IMP-007	Import shall report per-row outcomes and shall report explicitly when no rows were found.	MUST	DELIVERED
FR-IMP-008	Import shall tolerate mixed date formats within one workbook.	SHOULD	DELIVERED
FR-IMP-009	Import shall bound the parsed range so an oversized declared sheet range cannot hang the client.	MUST	DELIVERED

Table 8.4 — Import requirements. FR-IMP-009 addresses real production files declaring ranges near the spreadsheet row limit.

8.5 Task management

ID	REQUIREMENT	PRI.	STATUS
FR-TSK-001	Tasks shall be creatable against a shot or asset by users holding create authority.	MUST	DELIVERED
FR-TSK-002	Tasks shall be assignable to a user by holders of assignment authority.	MUST	DELIVERED
FR-TSK-003	Artists shall be able to claim an unassigned task for themselves.	SHOULD	DELIVERED
FR-TSK-004	Task status shall progress through the defined pipeline stages.	MUST	DELIVERED
FR-TSK-005	Tasks shall support threaded comments visible to all users who can see the task.	MUST	DELIVERED
FR-TSK-006	Commenting shall be available to observers without task-editing authority.	MUST	DELIVERED
FR-TSK-007	Tasks shall support checklists, attachments and dependencies.	SHOULD	DELIVERED
FR-TSK-008	Tasks shall be presentable in both list and board views.	SHOULD	DELIVERED
FR-TSK-009	Completed work shall be handed to the next department in pipeline order.	SHOULD	DELIVERED

Table 8.5 — Task management requirements. FR-TSK-006 exists so administrative oversight can comment without holding production authority.

8.6 Review and approval

ID	REQUIREMENT	PRI.	STATUS
FR-REV-001	Artists shall be able to upload media against a version of a shot or asset.	MUST	DELIVERED
FR-REV-002	Uploaded media shall be validated by type and served without inline rendering.	MUST	DELIVERED
FR-REV-003	The review player shall provide frame-accurate transport and scrubbing.	MUST	DELIVERED
FR-REV-004	The player shall provide freehand, arrow, rectangle and text annotation with colour selection and an eraser.	MUST	DELIVERED
FR-REV-005	The player shall provide A/B wipe, onion skinning, ghosting and side-by-side version comparison.	MUST	DELIVERED
FR-REV-006	Comments shall be anchored to a specific frame and version.	MUST	DELIVERED
FR-REV-007	The approval chain shall proceed artist, lead, production head, producer.	MUST	DELIVERED
FR-REV-008	Each approval transition shall record actor, action, timestamp and version.	MUST	DELIVERED
FR-REV-009	Approval transitions shall be refused to users lacking the corresponding capability, enforced server-side.	MUST	DELIVERED
FR-REV-010	A change request shall return the task to the artist with the reason recorded.	MUST	DELIVERED
FR-REV-011	A review queue shall show each user what awaits their action, and what is at other stages.	MUST	DELIVERED
FR-REV-012	A read-only presentation mode shall allow one presenter to drive a synchronised review.	COULD	DELIVERED
FR-REV-013	Users without review-submission authority shall be told why they cannot post, rather than shown a failing control.	SHOULD	DELIVERED

Table 8.6 — Review and approval requirements.

8.7 Client delivery

ID	REQUIREMENT	PRI.	STATUS
FR-CLI-001	Approved work shall be shareable with an external client by access link.	MUST	DELIVERED
FR-CLI-002	Client access shall require a code in addition to the link.	MUST	DELIVERED
FR-CLI-003	The access code shall be delivered to the client by email on share.	MUST	DELIVERED
FR-CLI-004	A client session shall expose only the items explicitly shared with it.	MUST	DELIVERED
FR-CLI-005	Client feedback shall be recorded against the shot and visible to the studio.	MUST	DELIVERED
FR-CLI-006	Client links shall be revocable.	MUST	DELIVERED

Table 8.7 — Client delivery requirements.

8.8 Time, attendance and reporting

ID	REQUIREMENT	PRI.	STATUS
FR-TIM-001	Production staff shall be clocked in automatically on sign-in and out on sign-out.	MUST	DELIVERED
FR-TIM-002	Presence shall reflect current activity rather than only an explicit sign-out.	MUST	DELIVERED
FR-TIM-003	Time shall be recordable against individual tasks.	MUST	DELIVERED
FR-TIM-004	Leads and production management shall see attendance and effort for staff within their scope.	MUST	DELIVERED
FR-TIM-005	Attendance data shall be visible only to those with a legitimate supervisory relationship.	MUST	DELIVERED
FR-TIM-006	A daily standup record shall be maintainable per user.	SHOULD	DELIVERED
FR-TIM-007	Analytics shall summarise throughput, workload and delivery risk.	SHOULD	DELIVERED

Table 8.8 — Time and reporting requirements.

8.9 Administration

ID	REQUIREMENT	PRI.	STATUS
FR-ADM-001	Administrators shall create, deactivate and reactivate accounts.	MUST	DELIVERED
FR-ADM-002	Administrators shall assign roles and departments.	MUST	DELIVERED
FR-ADM-003	The capability matrix shall be editable through the interface.	MUST	DELIVERED
FR-ADM-004	All significant changes shall be written to an audit log.	MUST	DELIVERED
FR-ADM-005	Audited changes shall be reversible where safe to do so.	SHOULD	DELIVERED
FR-ADM-006	Studio-level settings shall be configurable without redeployment.	SHOULD	DELIVERED
FR-ADM-007	Entity schemas and task templates shall be configurable.	COULD	DELIVERED
FR-ADM-008	Plug-in and integration surfaces shall be visible only to administrators until implemented.	SHOULD	DELIVERED

Table 8.9 — Administration requirements.

9 Non-functional requirements

9.1 Security

ID	REQUIREMENT	PRI.
NFR-SEC-001	Passwords shall be hashed with a memory-hard algorithm resistant to hardware-accelerated attack.	MUST
NFR-SEC-002	Session tokens shall carry a version that can be invalidated server-side before natural expiry.	MUST
NFR-SEC-003	Session validity checks shall fail closed if the validating store is unavailable.	MUST
NFR-SEC-004	Authentication, registration, password reset and password change shall each be independently rate limited.	MUST
NFR-SEC-005	Authentication responses shall not reveal whether an account exists.	MUST
NFR-SEC-006	Uploaded files shall be stored under generated names, never under client-supplied names.	MUST
NFR-SEC-007	Non-media uploads shall be served as downloads with content sniffing disabled.	MUST
NFR-SEC-008	File retrieval shall be scoped by tenant path and refuse cross-tenant access.	MUST
NFR-SEC-009	All database access shall be parameterised, with no string-concatenated queries.	MUST
NFR-SEC-010	Outbound email shall escape user-supplied content to prevent header or markup injection.	MUST
NFR-SEC-011	Rate limiting shall consume budget only on failed attempts, so that legitimate shared-network traffic is not locked out.	MUST

Table 9.1 — Security requirements. NFR-SEC-011 exists because studio staff share one network egress address; counting successful sign-ins would lock out the studio each morning.

9.2 Performance and capacity

ID	REQUIREMENT	PRI.
NFR-PRF-001	The system shall support at least 100 concurrent authenticated users on the reference hardware.	MUST
NFR-PRF-002	The tracking grid shall remain usable with at least 5,000 shot rows.	MUST
NFR-PRF-003	Frequently-read authorisation data shall be cached to avoid a database round trip per request.	MUST
NFR-PRF-004	Presence updates shall be throttled so that activity tracking does not generate a write per request.	MUST
NFR-PRF-005	Spreadsheet import shall bound its parsed range to protect against oversized declared ranges.	MUST
NFR-PRF-006	Media upload shall accept files up to 500 MB.	SHOULD

Table 9.2 — Performance and capacity requirements.

9.3 Availability, operability and maintainability

ID	REQUIREMENT	PRI.
NFR-AVL-001	The system shall run without dependency on outbound internet access for core function.	MUST
NFR-AVL-002	Services shall restart automatically after host or container failure.	MUST
NFR-AVL-003	Database schema migrations shall be applied automatically and idempotently on service start.	MUST
NFR-AVL-004	The database shall be backed up on a schedule, with restore verified.	MUST
NFR-AVL-005	Uploaded media shall persist independently of container rebuilds.	MUST
NFR-AVL-006	Cache loss shall degrade performance only, never correctness or session validity.	MUST
NFR-MNT-001	The codebase shall be statically typed end to end.	MUST
NFR-MNT-002	Application logs shall be structured and machine-parseable.	SHOULD
NFR-MNT-003	Errors shall be reported to a monitoring service with stack traces preserved.	SHOULD

Table 9.3 — Availability and maintainability requirements.

9.4 Usability and accessibility

ID	REQUIREMENT	PRI.
NFR-USE-001	Each role shall land on a default view appropriate to its responsibilities.	MUST
NFR-USE-002	An action a user cannot perform shall not be offered; where visibility is warranted, the reason shall be stated.	MUST
NFR-USE-003	Empty states shall indicate where relevant data can be found rather than terminating the path.	SHOULD
NFR-USE-004	The interface shall support both light and dark presentation.	COULD
NFR-USE-005	First-time users should receive guided orientation to the pages relevant to their role.	SHOULD
NFR-USE-006	Interactive controls shall be keyboard reachable with a visible focus state.	SHOULD

Table 9.4 — Usability requirements. NFR-USE-003 arose from a real incident: an empty default queue read as an absent feature.

10 Technology stack and decision register

Each significant technology choice recorded as a decision: what was chosen, what was rejected, on what criteria, and what consequence was accepted. Where the driver was a directive rather than a comparative study, this is stated plainly — a decision register that reconstructs post-hoc reasoning as though it were original analysis is not evidence, and any reviewer who checks will find it.

10.1 Stack summary

LAYER	SELECTED	PRINCIPAL ALTERNATIVES CONSIDERED
Language	TypeScript 5.9, end to end	JavaScript; Python or Go for the API tier
Package manager	pnpm 11 with workspaces	npm, Yarn
Build orchestration	Turborepo	Nx, Lerna, plain recursive scripts
Frontend	React with Vite	Next.js, Angular, Vue
Styling	Tailwind CSS v4	CSS modules, styled-components
Components	Radix UI primitives	Material UI, Ant Design, bespoke
Server state	TanStack Query	Redux Toolkit Query, SWR, bespoke
Client state	Zustand	Redux Toolkit, React Context
API framework	Express 5	NestJS, Fastify, tRPC
Validation	Zod, shared client and server	Joi, class-validator, manual
ORM	Prisma (migrated from Drizzle)	Drizzle, TypeORM, raw SQL
Database	PostgreSQL 15	MySQL, MongoDB, SQLite
Cache	Redis 7	In-process memory, Memcached
Password hashing	Argon2id	bcrypt, scrypt, PBKDF2
Sessions	Signed JWT in an HTTP-only cookie, with server-side version revocation	Server-side session store, opaque tokens
Logging	Pino, structured	Winston, console
Deployment	Docker Compose, five services	Kubernetes, bare-metal, managed PaaS
Reverse proxy	nginx	Caddy, Traefik, direct exposure

Table 10.1 — Technology stack summary. Decisions of consequence are recorded individually below.

10.2 Decision records

DEC-001 · TYPESCRIPT END TO END

Decision. A single language across frontend, API and shared packages, with types shared through workspace packages rather than duplicated.

Rejected. A Python or Go API tier, which would have required the request and response contract to be defined twice and kept in step by discipline alone.

Criteria. Contract safety across the network boundary; team size; the cost of a type mismatch reaching production.

Consequence accepted. Node's runtime characteristics for CPU-bound work are inferior to Go's. Accepted because the workload is I/O-bound: database reads, file serving and JSON.

DEC-002 · PRISMA AS THE ORM, REPLACING DRIZZLE

Decision. Prisma Client is the sole database access layer. The system was originally built on Drizzle ORM and migrated in full.

What the migration actually involved. Twenty-six tables declared across eight schema files, a custom migration runner, and query-builder calls in seventeen of twenty route files plus two library modules and three operational scripts. The work was planned as eleven sequenced tasks, executed against a clone restored from real production data, and cut over in a single planned downtime window rather than a gradual dual-write rollout.

Honest statement of the driver. The migration was undertaken on a directive to standardise on Prisma. It was not the outcome of a comparative trade-off study, and this document does not present one retrospectively.

Supporting technical justification. A single declarative schema file replaces eight hand-maintained definition modules; the generated client provides types derived from the schema rather than asserted alongside it; first-party migration tooling offers a deploy command suited to running unattended at container start, which is how migrations are applied here; and the larger installed base widens the pool of engineers who can maintain the system.

Constraints imposed on the migration. The schema was to remain byte-for-byte identical — an ORM swap, not a redesign. All business logic, including tenant scoping, capability gates, approval-stage gates and audit logging, was to be preserved with only query syntax changing. Every verification step was required to test over the same network path the real system uses.

Consequence accepted. Prisma's query engine adds a runtime dependency and its generated client must be produced during the container build; both were absorbed into the Dockerfile. Drizzle's thinner abstraction and closer-to-SQL query surface were given up.

ON THE VERIFICATION CONSTRAINT IN DEC-002

The requirement that every verification step test over the real network path was not a general principle but a response to a specific failure earlier the same day. A database password rotation had been verified by connecting to PostgreSQL from inside its own container, which passed — because local socket connections in that image use trust authentication and bypass password checks entirely. The real failure only appeared over the Docker network, after the change was live. Recording the incident alongside the decision is what stops the lesson being lost with the people who were present.

DEC-003 · POSTGRESQL AS THE DATABASE

Decision. PostgreSQL 15, self-hosted in a container with a scheduled backup sidecar.

Rejected. MongoDB, on the grounds that production tracking is inherently relational — a shot belongs to a sequence, belongs to an episode, belongs to a project, and integrity across those relationships is the product. MySQL was a viable alternative rejected on feature depth.

Consequence accepted. Operating a database is the studio's responsibility rather than a vendor's. Mitigated by NFR-AVL-004 and the backup service.

DEC-004 · REDIS AS CACHE ONLY, EXPLICITLY NOT DURABLE

Decision. Redis serves the authorisation cache, rate-limit counters and presence throttling. Persistence is deliberately disabled.

Rejected. Using Redis as a session store, which would have made session validity depend on cache durability.

Criteria. Cache loss must degrade performance, never correctness (NFR-AVL-006).

Consequence accepted. A restart empties the cache and the next requests fall through to the database. Verified in practice: an active session survives a full Redis restart, because validity derives from the signed token and the database, with Redis holding only a short-lived copy.

DEC-005 · ARGON2ID FOR PASSWORD HASHING

Decision. Argon2id.

Rejected. bcrypt, the common default, which is not memory-hard and is therefore more tractable to GPU-accelerated attack. PBKDF2 rejected on the same grounds.

Consequence accepted. Higher CPU and memory cost per authentication, which is the intended property.

DEC-006 · VERSIONED JWT SESSIONS RATHER THAN A SERVER-SIDE SESSION STORE

Decision. A signed token in an HTTP-only cookie carrying a version number, checked against the user's current version on every request.

Problem being solved. A plain stateless token cannot be revoked before expiry, so a logout, password change or offboarding left a valid token usable for its full lifetime. The version field restores revocation without giving up statelessness.

Consequence accepted. One validity lookup per request, mitigated by the cache in DEC-004, which fails closed (NFR-SEC-003).

DEC-007 · CAPABILITY-BASED AUTHORISATION RATHER THAN ROLE CHECKS

Decision. Permissions are named capabilities granted to roles in the database and checked per request.

Rejected. Hard-coded role comparisons in route handlers, which cannot be changed without a deployment and drift silently from what the interface displays.

Consequence accepted. An additional indirection between role and permission, and a matrix that must be administered. This is what makes BR-011 enforceable rather than conventional.

DEC-008 · DOCKER COMPOSE RATHER THAN KUBERNETES

Decision. Five containers on a single host: reverse proxy, API, database, cache, backup.

Rejected. Kubernetes, whose operational complexity is not justified for a single-studio deployment and which would require expertise the studio does not have.

Consequence accepted. No automatic horizontal scaling or multi-node failover. Acceptable at present scale and revisitable if a deployment outgrows one host.

DEC-009 · VITE AND REACT RATHER THAN NEXT.JS

Decision. A client-rendered single-page application built with Vite.

Rejected. Next.js, whose principal advantages are server-side rendering and search engine optimisation. Neither applies to an authenticated internal tool, and its server runtime would have added a second deployable component.

Consequence accepted. A larger initial bundle and no server rendering. Irrelevant on a studio local network.

DEC-010 · TURBOREPO FOR BUILD ORCHESTRATION

Decision. Turborepo over the pnpm workspace.

Criteria. Cached, parallel task execution, and attribution of a failure to a specific workspace package.

Measured outcome. A full workspace typecheck fell from approximately 17 seconds cold to 63 milliseconds fully cached, and an introduced type error was correctly isolated to the affected package while the others remained cached.

DEC-011 · RADIX UI PRIMITIVES RATHER THAN A COMPONENT LIBRARY

Decision. Unstyled accessible primitives, styled in-house with Tailwind.

Rejected. Material UI and Ant Design, both of which impose a visual identity that would have to be fought to achieve a studio-tool aesthetic, and both of which are heavier.

Consequence accepted. More styling work up front, in exchange for keyboard and screen-reader behaviour that would otherwise have to be built by hand (NFR-USE-006).

11 Constraints and assumptions

11.1 Constraints

CONSTRAINT	EFFECT ON THE SOLUTION
On-premise deployment	No dependency on managed cloud services; all state on studio hardware.
Restricted internet access	No runtime dependency on external content delivery networks or third-party APIs for core function.
Existing hardware	Single-host deployment sized to available capacity; no horizontal scaling assumption.
Shared network egress	All studio traffic presents one source address, so address-based protective measures must not treat the studio as one abusive client (NFR-SEC-O11).
Existing tracksheet formats	Import must adapt to the studio's file conventions; the studio will not adapt its files to the tool.
Live production data	Changes are made to a system holding real work in flight; migrations must be verified against a real-data clone before deployment.
Small engineering team	Favours managed complexity and conventional technology over bespoke infrastructure.

Table 11.1 — Constraints and their effect.

11.2 Assumptions

ID	ASSUMPTION	IF IT PROVES FALSE
ASM-1	Studio staff access the system from within the studio network or a trusted connection.	Public exposure would require transport encryption and hardened perimeter controls.
ASM-2	Production data is imported from existing tracksheets rather than re-entered.	Import effort becomes manual data entry; OBJ-5 fails.
ASM-3	One tenant per deployment in practice, though the model supports several.	Multi-tenant isolation would require independent penetration testing.
ASM-4	Media is uploaded in browser-playable formats.	A transcoding pipeline becomes necessary; currently out of scope.
ASM-5	An SMTP relay is reachable for notification and access-code delivery.	Client access codes must be communicated out of band.
ASM-6	Departments map to a linear pipeline order.	Branching pipelines would require the stage model to be extended.

Table 11.2 — Assumptions, each with its failure consequence.

12 Risks and mitigations

ID	RISK	LIKELI.	IMPACT	MITIGATION
RSK-1	Key-person dependency in engineering	High	High	This document set is the primary mitigation; maintain the SRS as a living record. Code carries decision rationale inline.
RSK-2	Data loss on the single host	Low	Critical	Scheduled backups with verified restore (NFR-AVL-004); media on a persistent volume independent of rebuilds.
RSK-3	Adoption failure — staff continue using spreadsheets	Medium	High	Import removes the migration barrier; role-appropriate landing views; guided orientation (NFR-USE-005).
RSK-4	Import fails on an unseen tracksheet layout	Medium	Medium	Tolerant column matching; unmatched data preserved; explicit reporting when nothing is recognised.
RSK-5	A protective measure locks out legitimate users	Medium	Medium	Failed attempts only consume rate-limit budget; thresholds set for shared-egress reality. Occurred in practice and was corrected.
RSK-6	Single host becomes a capacity ceiling	Low	Medium	Stateless API tier; horizontal scaling behind the existing proxy is available without redesign.
RSK-7	Verification passes for the wrong reason	Medium	High	Standing rule that verification must traverse the same path as real traffic. Derived from a live incident; see Section 10.2.
RSK-8	Client media exposed through a share link	Low	High	Links require a separate code, expose only shared items, and are revocable.

Table 12.1 — Risk register.

13 Dependencies

DEPENDENCY	TYPE	NATURE AND FAILURE BEHAVIOUR
Host server	Infrastructure	Runs all five containers. Total dependency; failure is an outage.
SMTP relay	External service	Password reset and client access-code delivery. Failure degrades those flows only; core tracking is unaffected.
Backup storage	Infrastructure	Destination for scheduled dumps. Failure is silent unless monitored — restore should be tested periodically.
Container runtime	Platform	Docker Engine and Compose on the host.
Browser	Client	Modern evergreen browser required for the review player's video and canvas features.
Error monitoring	External service	Optional. Absence removes visibility, not function.

Table 13.1 — External and infrastructure dependencies.

14 Acceptance and sign-off criteria

The conditions under which the sponsor accepts that this requirement set has been met.

ID	CRITERION	VERIFICATION
ACC-1	Every Must-priority functional requirement is delivered and demonstrable.	Demonstration
ACC-2	A production tracksheet imports without manual reformatting and without data loss.	Demonstration against a real file
ACC-3	Each role, signed in separately, sees only the data and controls its capability set permits.	Test per role
ACC-4	A shot completes the full chain from artist submission to client delivery with a complete recorded history.	End-to-end test
ACC-5	An external client reviews and returns feedback using only a link and code.	End-to-end test
ACC-6	A database backup restores to a working system.	Restore test
ACC-7	Security requirements NFR-SEC-001 to 011 are individually verified.	Inspection and test
ACC-8	The four documents in this set exist, are current, and are approved.	Inspection

Table 14.1 — Acceptance criteria.

15 Appendix A — Glossary

TERM	DEFINITION
Asset	A reusable production element — character, prop, environment — tracked in parallel to shots.
Capability	A named permission granted to a role and checked per request.
DCC	Digital Content Creation application, such as Maya, Blender, Nuke or Houdini.
Episode	A single instalment of an episodic production, containing sequences.
Onion skinning	Display of adjacent frames semi-transparently to judge motion between them.
Pipeline stage	An ordered department position through which a shot passes.
Playblast	A fast preview render used for review rather than final output.
Retake	Work redone following rejection or a change request.
Sequence	A group of related shots within an episode.
Shot	The atomic unit of production work — a continuous run of frames.
Tenant	An isolated studio instance within the data model.
Tracksheet	The spreadsheet a studio uses to track shot progress.
Version	A numbered iteration of deliverable media for a shot or asset.
A/B wipe	A draggable divider comparing two versions in the same frame.

Table 15.1 — Glossary of production and system terms.

16 Appendix B — Traceability seed

Business requirement to objective to functional requirement. The SRS extends this matrix to individual test cases; maintaining it in both directions is what makes the requirement set auditable.

BUSINESS REQ.	OBJECTIVE	FUNCTIONAL REQUIREMENTS
BR-001	OBJ-1	FR-TRK-001 ... 007, FR-IMP-001 ... 009
BR-002	OBJ-3	FR-REV-004, FR-REV-006, FR-REV-008
BR-003	OBJ-4	FR-REV-007 ... 010, FR-ADM-004
BR-004	OBJ-5	FR-IMP-001 ... 009
BR-005	OBJ-6	FR-REV-002, NFR-SEC-006 ... 008, NFR-AVL-001
BR-006	OBJ-2	Satisfied by deployment model; no per-seat mechanism exists in the system
BR-007	OBJ-1	FR-RBAC-001 ... 008, FR-TIM-005
BR-008	—	FR-CLI-001 ... 006
BR-009	OBJ-7	FR-TIM-001 ... 007
BR-010	OBJ-6	NFR-AVL-001 ... 006
BR-011	—	FR-RBAC-004, FR-TSK-006, FR-REV-013
BR-012	—	FR-AUTH-003, FR-AUTH-010 ... 013
BR-013	—	FR-ADM-004, FR-ADM-005
BR-014	—	FR-TIM-004, FR-TIM-007
BR-015	—	FR-ADM-008 (surface only; deferred)
BR-016	—	Out of scope — see Table 4.2

Table 16.1 — Traceability seed matrix.

MAINTAINING THIS MATRIX

A requirement added to the SRS without a corresponding entry here indicates either a missing business requirement or scope that was never agreed. Both are worth catching at review rather than at acceptance.